Student housing in City of Brussels - iterative decision

Illustrative office conversion: 120 rooms, including 24 social rooms in final design. All prices and project-specific feasibility are assumptions.

Municipal distinction: City of Brussels publishes 2021 guidance for 20% social student housing from 30 created units. Current guidance listing checked; exact permit conditions and university agreement remain outstanding. No corresponding universal regional quota or different Ixelles threshold has been verified.

The statement that Brussels is Belgium’s most expensive student city was not independently established; it is not used as a calculation input.

Three evaluated model iterations

{"iteration": "iteration_1", "social_rooms": 12, "required_quota": 0.1, "acquisition": 3000000, "works": 2200000, "NPV": 1407448.3116851114, "loss_probability": 0.326, "NPV_P10": -805803.7673382424, "equity_funded": 3541715.0}

{"iteration": "iteration_2", "social_rooms": 24, "required_quota": 0.2, "acquisition": 3000000, "works": 2200000, "NPV": 1130336.3875134084, "loss_probability": 0.394, "NPV_P10": -887018.5419626865, "equity_funded": 3541715.0}

{"iteration": "iteration_3", "social_rooms": 24, "required_quota": 0.2, "acquisition": 2211539.073810945, "works": 2200000, "NPV": 2017354.9294760954, "loss_probability": 0.1, "NPV_P10": 4.602043190971017e-10, "equity_funded": 3098205.7290186565}

Iteration 1 is preliminary and not compliant with the adopted 20% design requirement. Iteration 2 reserves 20%. Iteration 3 changes only acquisition price to improve downside risk, preserving housing cost ceilings.

The 90% probability target and triangular shock distributions are investor assumptions. Reusing the same seeded draws makes model changes comparable, not empirically certain.

Decision and limitations

Conditional negotiated acquisition ceiling: EUR 2,211,539. Final NPV EUR 2,017,355; simulated loss probability 10.0%.

Do not commit without a vendor agreeing to the price, verified conversion costs, planning permission, university/social allocation terms and loan underwriting.

The equity amount in the iteration ledger is a prospective funding requirement. No real capital was invested or reinvested.

Inputs are classified as constants, controls and uncertain inputs, with approximate versus inherent uncertainty. Sensitivity, risk distributions and iteration charts support decisions.

Pre-tax financial model; unknown exact transaction VAT/tax treatment, static rent-occupancy relation, and unverified physical capacity limit confidence.

References

https://www.bruxelles.be/sites/default/files/bxl/Recommandations_logements_etudiants_VBX_17juin2021.pdf

https://www.vub.be/en/studying-vub/practical-info-for-students/student-housing/practical-housing-information

https://www.brukot.be/en/student-guide/rent-stats

https://be.brussels/en/taxes-finances/taxes/property-related-taxes/registration-duties

https://be.brussels/fr/logement/location/conformite-logement/normes-et-securite

https://be.brussels/fr/logement/location/bail-dhabitation/loyers-de-reference

https://ibsa.brussels/themes/logement/prix-des-logements

Current municipal guidance listing: https://www.bruxelles.be/plans-reglements-urbanisme